

LOW ACRYLONITRILE NBR: PATENT LANDSCAPE FOR COMPOSITION CONTROL, SOLUTION POLYMERIZATION, AND LOW-MOONEY RUBBER DESIGN

1. ABSTRACT

This report synthesizes three primary patents relevant to low acrylonitrile nitrile-butadiene rubber (NBR) and closely related low-viscosity nitrile rubber systems. The target is defined as NBR grades with reduced acrylonitrile content and/or composition control strategies that enable low-ACN elastomers with manageable processability, while avoiding hydrogenated, carboxylated, or otherwise excluded variants unless they are directly relevant as downstream analogs. Across the selected patents, the landscape spans three distinct but related themes: (1) low-Mooney nitrile rubber generated by metathesis followed by hydrogenation, (2) solution-phase free-radical NBR synthesis in organic solvents using modifier/RAFT-like control concepts, and (3) swellable rubber formulations that incorporate nitrile elastomers, including hydrogenated nitrile elastomers, as part of a high-performance downhole composition.

EP1554342B1 focuses on a two-step route to low-Mooney hydrogenated nitrile rubber, where a statistical butadiene-acrylonitrile terpolymer is first metathesized in monochlorobenzene under ethylene pressure and then hydrogenated with Wilkinson's catalyst at elevated temperature and hydrogen pressure. The disclosed feedstock includes 47 mol% butadiene and 21 mol% acrylonitrile, showing that low-ACN nitrile backbones can be converted into highly processable low-viscosity HNBR. EP2478021B1 discloses solution polymerization of nitrile rubbers in organic solvents, using acrylonitrile, 1,3-butadiene, and modifiers/initiator systems to control molecular weight and enable direct synthesis of NBR in solution rather than conventional emulsion polymerization. It emphasizes solvent choice, regulator concentration, conversion control, and post-polymerization coagulation. US10385647B2 is not a synthesis patent for NBR itself, but it is relevant because it specifies swellable rubber compositions containing nitrile elastomer and butadiene rubber, with explicit ranges for hydrogenated nitrile elastomer hydrogenation, Mooney viscosity, and SAP loading; it demonstrates how nitrile-based elastomers are formulated for high-temperature, high-pressure service.

Overall, the evidence suggests that low-ACN NBR synthesis is best approached through careful monomer feed control in solution or emulsion polymerization, with downstream viscosity tailoring

via molecular-weight control or metathesis/hydrogenation when HNBR analogs are acceptable. The most actionable process trends are solvent-based polymerization, controlled initiator/modifier use, and explicit management of Mooney viscosity and conversion to obtain processable nitrile rubber grades.

2. METHODOLOGY

PRIMARY PATENT EVIDENCE

Patent 1

- Patent Number: EP1554342B1
- Patent Title: Polymer blends comprising low molecular weight nitrile rubber
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2013
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses a two-step route to low-Mooney hydrogenated nitrile rubber: first a metathesis reaction of a nitrile rubber cement in monochlorobenzene under ethylene, followed by hydrogenation in the same reactor using Wilkinson's catalyst and triphenylphosphine. The starting nitrile rubber is a statistical butadiene-acrylonitrile terpolymer, and the process is used to generate low-viscosity HNBR suitable for blending into multimodal polymer systems. - Cement concentration: 6% — Example 1a - Co-olefin: ethylene — Example 1a - Co-olefin concentration: 3447 kPa [500 psi] — Example 1a - Agitator speed: 600 rpm — Example 1a - Reactor temperature: 80°C — Example 1a - Catalyst loading: 0.5 phr — Example 1a - Solvent: monochlorobenzene — Example 1a - Butadiene content: 47 mol% — Example 1a substrate - Acrylonitrile content: 21 mol% — Example 1a substrate - Butylacrylate content: 32 mol% — Example 1a substrate - Mooney viscosity ML(1+4)@100°C: 29 — Example 1a substrate - Ethylene degassing pressure: 689 kPa [100 psi] — Example 1a - Reaction time: 1 hours — Example 1a - Hydrogen pressure: 8274 kPa [1200 psi] — hydrogenation conditions - Hydrogenation temperature: 138°C — hydrogenation conditions - Agitator speed: 600 rpm — hydrogenation conditions - Wilkinson's catalyst loading: 0.08 phr — hydrogenation conditions - Triphenylphosphine: 1 phr — hydrogenation conditions - Hydrogenation time: 4 hours — Example 1a compound 1a - Hydrogenation time: 3 hours — Example 1a compound 1b - Residual double bonds hydrogenated: 99.6% — Example 1a compound 1a - Residual double bonds hydrogenated: 99.3% — Example 1a compound 1b - Acrylonitrile content: 34 mol% — Example 2a-f HNBR blend component - Residual double bond content: <0.9% — Example 2a-f HNBR blend component - Mooney viscosity ML (1+4)@100°C: 70 MU — Example 2a-f HNBR blend component - Total solid content: 6% — Example 2a-f

Relevant Experimental Evidence - Example 1a: low-Mooney nitrile rubber was prepared by metathesis at 80°C and 500 psi ethylene, then hydrogenated at 138°C and 1200 psi H₂ to >99% residual double bond saturation. - Example 1a: the starting terpolymer contained 21 mol% acrylonitrile, demonstrating relevance to low-ACN nitrile feedstocks. - Example 2a-f: the low-Mooney hydrogenated rubber was blended with a higher-Mooney HNBR to create multimodal molecular-weight distributions. - Example 2a-f: the blend concept shows how low-viscosity nitrile rubber can be combined with higher-viscosity HNBR for processability tuning.

Technical Relevance This patent is highly relevant because it directly addresses low-Mooney nitrile rubber derived from a nitrile-containing precursor with explicit acrylonitrile content. Although the final product is hydrogenated, the disclosed feed composition and viscosity-control strategy are useful for designing low-ACN NBR/HNBR precursor systems.

Patent 2

- Patent Number: EP2478021B1
- Patent Title: Nitrile rubbers and production of same in organic solvents
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2015
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses free-radical preparation of nitrile rubbers in organic solvent rather than conventional emulsion polymerization. Acrylonitrile and 1,3-butadiene are polymerized in the presence of an organic solvent and modifier/regulator system, with oxygen exclusion, controlled initiator dosing, and post-polymerization coagulation/isolation of the polymer. - Vazo® 88: 284.9 mg (1.17 mmol, 0.31 phm) — Example 1 - DoPAT: 204 mg (0.583 mmol, 0.22 phm) — Example 1 - Dimethylacetamide: 50 ml (51 phm) — Example 1 - Acrylonitrile: 43 ml (653.3 mmol, 37 phm) — Example 1 - Argon degassing: 10 minutes — Example 1 - 1,3-Butadiene: 90 ml (1077 mmol, 63 phm) — Example 1 - Pressure: 3 bar — Example 1 - Reaction temperature: 100°C — Example 1 - Polymerization time: 9 hours — Example 1 - Final conversion: 39% — Example 6 - Regulator amount: 2.22 phm — Example 7 - Polymerization time: 9 h — Example 7 - Final conversion: 48% — Example 7 - Styrene: 2.79 g (26.8 mmol, 50 phm) — Example 8 - Reaction temperature: 100°C — Example 8 - Reaction time: 13 hours — Example 8 - Final conversion: 9.2% — Example 8 - Vazo® 88 post-dosing: 128 mg (0.523 mmol, 0.15 phm) — Examples 18, 20, 23 - Post-dosing time: 2.5 and 6 hours — Examples 18, 20, 23 - Reaction temperature: 100°C — Examples 17-23 - Pressure: 3 bar — Examples 17-23 - Reaction time: 9 hours — Examples 17-23

Relevant Experimental Evidence - Example 1: NBR was synthesized in DMAc from acrylonitrile and butadiene using Vazo® 88 and DoPAT, with polymerization initiated at 100°C under 3 bar butadiene. - Example 6: a terpolymer containing hydroxyethyl acrylate was prepared by the same

solution route and stopped at 39% conversion after 6 hours. - Example 7: increasing regulator amount to 2.22 phm produced a deliberately low-molecular-weight NBR and 48% conversion after 9 hours. - Example 8: the low-MW NBR from Example 7 was chain-extended with styrene to form an NBR-block-styrene terpolymer. - Examples 17-23: solution polymerization was repeated with HDODA and/or HEA monomers, with 100°C, 3 bar, and 9 h as the common process window.

Technical Relevance This patent is relevant because it provides a direct solution-polymerization route for NBR with explicit acrylonitrile/butadiene feed control and molecular-weight regulation. Such a route is central to low-ACN NBR synthesis when solvent-based composition and viscosity control are desired.

Patent 3

- Patent Number: US10385647B2
- Patent Title: Swellable rubber compositions
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2019
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent does not disclose NBR synthesis; instead, it discloses swellable rubber compositions for packers that include a nitrile elastomer, butadiene rubber, sulfur curing agent, silica, and powder super absorbent polymer. The nitrile elastomer may be a hydrogenated nitrile elastomer with defined hydrogenation and Mooney viscosity ranges, and the compositions are cured and tested under brine, temperature, and pressure conditions. - Powder super absorbent polymer: about 125 Phr to about 175 Phr — Claim 2 - Powder super absorbent polymer: about 170 Phr to about 210 Phr — Claim 3 - Powder super absorbent polymer: at least about 150 Phr, about 175 Phr, about 200 Phr or about 225 Phr — Claim 4 - Powder SAP fraction: at least about 50%, 60%, 70%, 80%, 90%, or 95% by weight of total SAP — Claim 5 - Hydrogenation: about 70 percent to about 91 percent — Summary - Mooney viscosity ML 1+4 (100°C): about 47 to about 72 — Summary - Percent unsaturation: about 9 percent to about 30 percent — Summary - Swelling: at least about 50%, 60%, 65%, 70%, 75%, 80%, or 90% by volume — Summary - Swelling time: about 96 hours — Summary - Brine concentration: about 5% — Summary - Temperature: about 200 degrees Fahrenheit — Summary - Temperature: about 275 degrees Fahrenheit — Summary - Pressure: at least about 3000, 4000, 5000 or 6000 psi — Summary - Curing temperature: 330 degrees Fahrenheit — Example 1 - Curing pressure: 70 psi — Example 1 - Curing time: 2.5 hours — Example 1 - Test temperature: 275 degrees Fahrenheit — Example 1 - Test salinity: 16% brine — Example 1 - Differential pressure: at least 5000 psi — Example 1 - Base pipe OD: 4.5 inches — Example 1 - Packer OD: 5.7 inches — Example 1 - Element length: 5 feet — Example 1 - NBR rubber loading: 10% — Example 4 - HNBR loading: 10% — Example 4 - 96 hour swell: 75.8% — Example 4 (10% DN3350, 200°F, 5% brine) - 96 hour swell: 77.8% — Example 4 (10% HNBR, 200°F, 5% brine)

Relevant Experimental Evidence - Example 1: a swellable packer using Composition B was cured at 330°F and 70 psi for 2.5 hours and then held in 16% brine at 275°F under pressure. - Example 1: the packer withstood at least 5000 psi differential pressure after multi-day exposure. - Example 2: SAP particle size and loading were compared, showing strong dependence of swell and leaching on SAP morphology and loading. - Example 3: powder SAP loadings were projected to achieve 75% swell, with about 205 Phr predicted for powder SAP. - Example 4: 10% NBR and 10% HNBR formulations both showed strong swell performance, with HNBR offering improved thermal stability potential.

Technical Relevance This patent is relevant as a formulation and application reference for nitrile-based elastomers, including hydrogenated nitrile elastomers, in high-performance swellable systems. It does not advance low-ACN synthesis directly, but it provides useful downstream composition constraints and processing windows for nitrile rubber grades intended for demanding service.

3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

- EP1554342B1 and EP2478021B1 both address nitrile-rubber composition control, but EP1554342B1 uses a low-Mooney precursor followed by metathesis/hydrogenation, whereas EP2478021B1 uses direct solution free-radical polymerization of acrylonitrile/butadiene in organic solvent.
- EP1554342B1 explicitly discloses a starting terpolymer with 21 mol% acrylonitrile, while EP2478021B1 discloses acrylonitrile feed at 37 phm and butadiene at 63 phm in Example 1; both are consistent with low-to-moderate ACN NBR precursor design rather than high-ACN grades.
- EP1554342B1 relies on ethylene pressure during metathesis and hydrogen pressure during hydrogenation; EP2478021B1 relies on 3 bar monomer pressure, 100°C polymerization temperature, and modifier/initiator control in solution.
- EP1554342B1 emphasizes Mooney viscosity reduction to below 10 for the hydrogenated low-Mooney component, while EP2478021B1 emphasizes molecular-weight control through regulator concentration and conversion management to obtain processable NBR.
- EP2478021B1 uses organic solvent polymerization with DMAc and post-polymerization coagulation, whereas EP1554342B1 uses monochlorobenzene for both metathesis and hydrogenation; both are solvent-based but serve different process objectives.
- US10385647B2 is formulation-focused rather than synthesis-focused, but it confirms that nitrile elastomers with about 70-91% hydrogenation and Mooney viscosity about 47-72 are suitable in swellable packer compositions alongside powder SAP.
- Across the set, the most explicit low-ACN composition evidence is in EP1554342B1 (21 mol% acrylonitrile substrate), while the most explicit direct NBR synthesis evidence is in EP2478021B1.

- The patents collectively suggest that low-ACN NBR development benefits from controlled monomer feed ratios, solvent-based polymerization, and downstream viscosity tailoring when hydrogenated analogs are acceptable.

4. CONCLUSION

The evidence indicates that low acrylonitrile NBR development is best supported by solvent-based polymerization with explicit monomer-ratio control and, where needed, downstream molecular-weight reduction or hydrogenation to improve processability. EP2478021B1 provides the clearest direct synthesis route for NBR in organic solvent, while EP1554342B1 demonstrates how a nitrile-containing precursor with 21 mol% acrylonitrile can be converted into a low-Mooney rubber via metathesis and hydrogenation. US10385647B2 is not a synthesis patent, but it confirms practical formulation windows for nitrile elastomers in demanding applications. Based on the evidence, recommended synthesis parameters for low-ACN NBR are controlled acrylonitrile/butadiene feed ratios, solution polymerization in an oxygen-free organic solvent, temperature near 100°C for radical polymerization, and viscosity control via regulator or post-polymerization modification when low-Mooney performance is required.

5. REFERENCES

- EP1554342B1 | Polymer blends comprising low molecular weight nitrile rubber | Not disclosed | EP | 2013 | <https://patents.google.com/patent/EP1554342B1/en>
- EP2478021B1 | Nitrile rubbers and production of same in organic solvents | Not disclosed | EP | 2015 | <https://patents.google.com/patent/EP2478021B1/en>
- US10385647B2 | Swellable rubber compositions | Not disclosed | US | 2019 | <https://patents.google.com/patent/US10385647B2/en>